AYAN PAL

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INTERNSHIP

FlyRank AI (Machine Learning Intern) **Jun'26 – Aug'26**

Problem	• The marketing team managed **~30,000 existing web pages** and could not audit each manually; built a data-driven content prioritization system to decide which pages to refresh first.
Initiatives	• Performed **EDA** and **feature engineering** on **44 features** (impressions, clicks, CTR, average position, content age, word count, engagement metrics) using Python, Pandas, Scikit-learn. • Defined a declining-page target and excluded all target-defining variables from inputs to prevent data leakage. • Built an interpretable r**ule-based baseline model** to benchmark ML. • Identified impressions, average position, content age and word count as key predictive signals. • Trained and compared **Logistic Regression, Decision Tree and Random Forest** using a **client-based holdout** to test generalization; optimized for **Precision@50** since the team can review only a few pages.
Impact	• Random Forest achieved **0.74 Precision@50 vs 0.24 for the baseline (~3x lift)** and ROC-AUC of 0.75 vs 0.63. • Random Forest beat Decision Tree (0.54) and Logistic Regression (0.40) on Precision@50. • Delivered a ranked, explainable refresh queue (score, reasons, action), turning a 30K-page audit into a **top-50 list**.

ACADEMIC QUALIFICATIONS

Degree / Examination	Institution	CGPA / %	Year
B.Tech (Engineering Physics)	Delhi Technological University (DTU), New Delhi	8.162 CGPA	2023–2027
Class XII, CBSE	DAV Public School, East of Loni Road, Delhi	86%	2021–2022
Class X, CBSE	DAV Public School, East of Loni Road, Delhi	88%	2019–2020

PROJECTS

Ola Ride-Hailing Analytics [GitHub]	• Built an **end-to-end analytics project on ride-booking data** (ride volume, booking status, vehicle-type performance, payment-method revenue split, top customers) using SQL (joins, CTEs, aggregations). • Validated SQL outputs with Excel pivot tables and formulas; built an interactive **Power BI dashboard** with KPI cards, slicers and drill-throughs. • Split cancellations into customer-side vs driver-side; found **Sedan cancellations at 16% vs 11% overall** (driver-side 9% vs 5%) alongside a 15% Sedan booking-value dip; recommended driver-availability monitoring.
Walmart Sales Data Analysis [GitHub]	• Built an end-to-end ETL pipeline: Kaggle API → Pandas cleaning (nulls, duplicates, data types, derived Total Amount) → PostgreSQL via SQLAlchemy on **10,000+ transactions**. • Wrote **9+ advanced SQL queries** using GROUP BY, aggregations and RANK() window functions (top category, busiest day, profit). • Flagged a **12% YoY branch revenue decline ($50M to $44M)**; drilled down by category and time of day to trace it to afternoon fresh-produce sales drops; proposed checks on inventory, staffing and footfall.
Restaurant Recommendation System [GitHub]	• Performed **EDA**, **data cleaning** and **NLP preprocessing** (stop-word, URL, punctuation removal) on Delhi reviews. • Vectorized reviews with **TF-IDF (1–2 word n-grams)** and used cosine similarity to build a content-based recommendation system. • Combined similarity candidates with cuisine, mean rating and cost to output a ranked top 10 list of similar restaurants.

POSITIONS OF RESPONSIBILITY

Let's Talk DTU (Head of Design) **Aug'23 – Dec'25**

Leadership	• Directed data-informed creative strategy across **10+ campaigns**, analyzing reach and engagement metrics to drive a **documented 20% boost in audience engagement**. • Co-led a design vertical of contributors, streamlining briefs, feedback loops and deadlines to deliver **50+ digital and print assets** with **100% on-time delivery**.

EXTRA-CURRICULAR ACHIEVEMENTS

Hackathons	• Selected among the **Top 10 teams at Hack4Delhi** organized by IEEE NSUT for a secure e-voting system; owned the backend and security layer (PostgreSQL, encryption, BB84-style QKD to detect eavesdropping).
Competitive Programming	• **LeetCode rating 1506** and **CodeChef rating 1246**; solved **300+ problems** across platforms , solid problem-solving base.

TOOLS & SKILLS

Languages & DB	SQL , PostgreSQL, Python, C++, SQLAlchemy
Analytics & ML	Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, TensorFlow, Keras, EDA, Feature Engineering, Statistical Analysis, Data Cleaning & Wrangling, ETL Pipelines, Model Evaluation, NLP (TF-IDF)
BI & Tools	Power BI, Microsoft Excel , Tableau, Dashboarding, KPI Tracking, Git
Product Analytics	A/B Testing, Funnel & Cohort Analysis, User Retention / Churn, DAU/MAU, LTV/CAC
Business Analysis	Root Cause Analysis (RCA), BRD/FRD, User Stories, Gap Analysis, GTM strategy